

Assignment 1	
Example	
25.2	26.1
25.3	26.2
25.4	26.3
	26.4
Problem 1	

Quiz
From start to Gauss's Law

Potential energy:

- Potential energy (P.E)
- Gravitational potential energy (G.P.E)
- Electrical Potential energy (E.P.E)
- Newton's Law of gravitation.

$$F = \frac{GM_1M_2}{r^2}$$

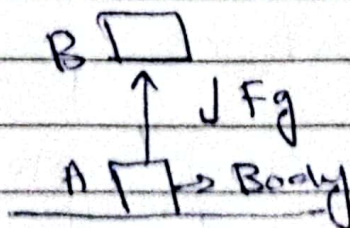
- Inverse square law
- Attractive force
- It exist between two masses.
- Its field is called gravitational force

$$g = \frac{F}{m} \because F = ma \rightarrow F = mg$$

- It is a conservative force.
- Work is independent on path followed.

G.P.E

Work done on an object in moving it from one point to another point against gravity:



E.P.E

Coulomb's law

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

→ Attractive / Repulsive

→ $E = \frac{F}{q}$ ⇒ Force per unit charge.

→ it exists b/w two charges

→ It is a conservative force

Definition of E.P.E

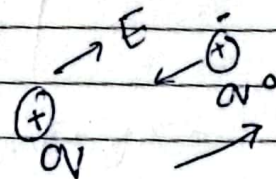
Work done on a charge in moving it from one point to another point against the direction of electric field.

$$\Delta U = -W$$

It increases the energy of charge

Electric potential energy,

Let us consider two points A and B around a charge q_1 , such that distance of point A from charge is r_A and distance of point B from charge is r_B . We consider another charge q_2 is at distance ' r ' from ' q_1 ' so its E.P.E is:



$$\Delta U = -W$$

$$\Delta U = - \int_{r_a}^{r_b} \vec{F} \cdot d\vec{s}$$

Suppose
 $ds = dr$

$$\Delta U = - \int_{r_a}^{r_b} F \cdot dr \cos 0$$

$$\Delta U = - \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} dr$$

$$\Delta U = - \frac{1}{4\pi\epsilon_0} q_1 q_2 \int_{r_a}^{r_b} \frac{1}{r^2} dr$$

$$\Delta U = - \frac{1}{4\pi\epsilon_0} q_1 q_2 \int_{r_a}^{r_b} r^{-2} dr$$

$$\Delta U = - \frac{1}{4\pi\epsilon_0} q_1 q_2 \left[\frac{r^{-2+1}}{-2+1} \right]_{r_a}^{r_b}$$

$$\Delta U = - \frac{1}{4\pi\epsilon_0} q_1 q_2 \left[\frac{r^{-1}}{-1} \right]_{r_a}^{r_b}$$

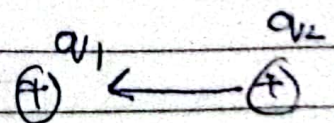
$$\Delta U = \frac{1}{4\pi\epsilon_0} q_1 q_2 \left[\frac{1}{r} \right]_{r_a}^{r_b}$$

$$= \frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{r_a}^{r_b}$$

$$\Delta U = \frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right] \rightarrow \text{Final Expression for electric P.E}$$

Case 1:

if q_2 moves towards q_1
 $r_b < r_a$



it will move against field so its energy will be increased. $\Delta U > 0$

Case 2:

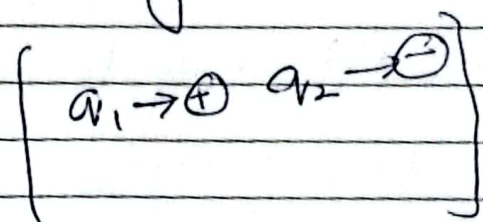
if q_2 moves away from q_1 ,
it will move in the direction of $\left[\begin{array}{cc} \leftarrow \oplus_{q_1} & \oplus_{q_2} \rightarrow \end{array} \right]$

E.F so its energy will decrease $\Delta U < 0$

Case 3:

if both charges have opposite signs.

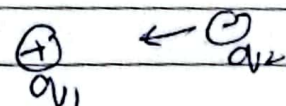
$$\Delta U = \frac{-q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$



* if q_2 moves towards q_1

$$r_b < r_a$$

$$\Delta U < 0$$

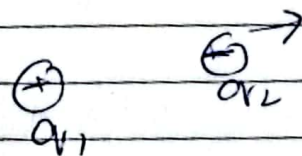


So its energy will decrease.

* if q_2 moves away from q_1

$$r_b > r_a$$

$$\Delta U > 0$$

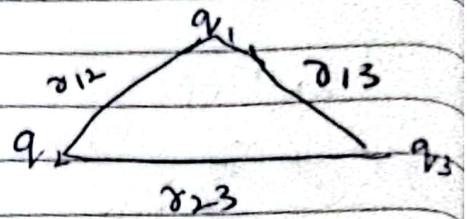


Electric Potential energy for System of charges:

Let us consider we have 3

charges q_1 , q_2 and q_3 and

distance b/w them is r_{12} , r_{23} , r_{13}



→ E.P.E due to q_1 and q_2

$$U_1 = \frac{k q_1 q_2}{r_{12}}$$

→ E.P.E due to q_2 and q_3

$$U_2 = \frac{k q_2 q_3}{r_{23}}$$

→ E.P.E due to q_1 and q_3

$$U_3 = \frac{k q_1 q_3}{r_{13}}$$

In order to calculate the total energy, we will add up the energies of charges.

$$U(r) = U_1 + U_2 + U_3$$

$$U(r) = \frac{k q_1 q_2}{r_{12}} + \frac{k q_2 q_3}{r_{23}} + \frac{k q_1 q_3}{r_{13}}$$

$$U(r) = k \frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}}$$

$$U(r) = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right] \quad \text{Total energy due to system of charges}$$

Electric Potential:

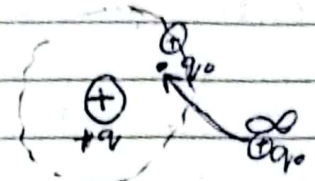
Electric potential at a point is

defined as amount of work done

In bringing a small unit positive

charge from infinity upto that

point by keeping it in electrostatic equilibrium



فيلزم بآبارك الله
في كل وقت
infinite

$$\rightarrow \text{Unit} \rightarrow V = \frac{W}{q_0} = \frac{J}{C} = 1 \text{ Volt} = 1V$$

for opposite charges ~~not~~ in electrostatic equilibrium

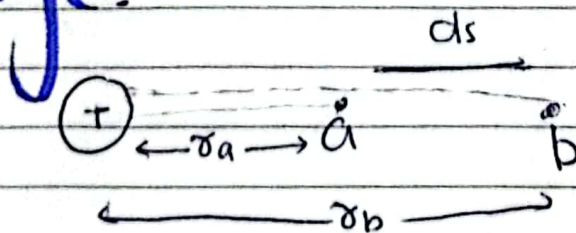
$$V = \frac{Q}{q_0}$$

$$V = 0$$

speed should be
constant $\rightarrow$ Electrostatic
equilibrium

$\rightarrow$ It is a scalar quantity.

Electric Potential due to point charge:



let us consider two points a and b around
a positive charge q' such that distance
of point a from q' is r_a and distance of
point b from q' is r_b

work done in moving charge ' q_0 ' from point 'b' to point 'a':

$$W_{b \rightarrow a} = \vec{F} \cdot \vec{ds}$$

integrate it

$$W_{b \rightarrow a} = \int_b^a \vec{F} \cdot \vec{ds} \quad \{ds = dr\}$$

$$W_{b \rightarrow a} = \int_{r_b}^{r_a} F dr \cos \theta \quad \{\cos \theta = 180\}$$

$$W_{b \rightarrow a} = \int_{r_b}^{r_a} F dr (-1)$$

$$W_{b \rightarrow a} = - \int_{r_b}^{r_a} F dr$$

$$W_{b \rightarrow a} = - \int_{r_b}^{r_a} q_0 E dr$$

$$\left\{ E = \frac{q}{r^2} \right\}$$

$$W_{b \rightarrow a} = -q_0 \int_{r_b}^{r_a} \frac{k q}{r^2} dr$$

$$W_{b \rightarrow a} = \frac{-q_0 q}{4\pi\epsilon_0} \int_{r_b}^{r_a} \frac{1}{r^2} dr$$

$$W_{b \rightarrow a} = \frac{-q_0 q}{4\pi\epsilon_0} \int_{r_b}^{r_a} r^{-2} dr$$

$$W_{b \rightarrow a} = \frac{-q_0 q}{4\pi\epsilon_0} \left[\frac{r^{-2+1}}{-2+1} \right]_{r_b}^{r_a}$$

$$W_{b \rightarrow a} = \frac{q_0 q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{r_b}^{r_a}$$

$$W_{b \rightarrow a} = \frac{q_0 q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_b} \right]$$

$$\frac{W_{ba}}{q_0} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_b} \right]$$

$$\Delta V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_b} \right]$$

$$V_a - V_b = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_b} \right]$$

if point b is at infinity

$$V_b = V_\infty = 0$$

$$\frac{1}{r_b} = \frac{1}{\infty} = 0$$

$$\Rightarrow V_a = \frac{q}{4\pi\epsilon_0} \frac{1}{r_a}$$

In general, we can write,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \rightarrow \text{electric potential at a point due to point charge}$$

Electric Potential due to N-charges

Let us consider N-number charges

$q_1, q_2, q_3, \dots, q_n$

We want to calculate the ~~P.O.E~~ ^{E.P} at

Point P due to this charges

Let the distance from q_1 is r_1
 q_2 is r_2
 q_n is r_n

In order to calculate the total electric Potential

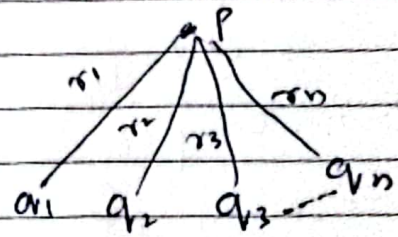
$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1}$$

$$V_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

...

$$V_n = \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_n}$$

Total Electric Potential is

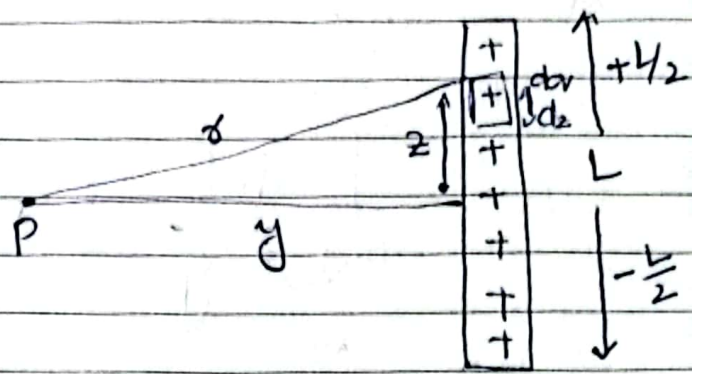


$$V_1 + V_2 + V_3 + \dots + V_n = V$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} + \frac{1}{4\pi\epsilon_0} \frac{q_3}{r_3} + \dots + \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_n}$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots + \frac{q_n}{r_n} \right]$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$



{ Conductor
uniformly
distributed
charge }

Electric potential due to line of charge:

Let us consider a +ve charge is uniformly distributed over a line of length 'L'. So we want to calculate E.P at point 'P' which is at distance 'y' from the centre of rod.

Consider a small charge 'dq' such that linear charge density will be $\lambda = \frac{dq}{dz} \Rightarrow dq = \lambda dz$

E.P due to small charge

dq

$$V = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

Total E.P:

In order to calculate E.P due to line of charge (entire charge) we will integrate

{ Linear charge
density
 $\lambda = \frac{q}{L}$
 $L = \frac{q}{\lambda}$
 $q = \lambda L$ }

$$\int dv = \int_{-L/2}^{+L/2} \frac{1}{4\pi\epsilon_0} \frac{da}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_{-L/2}^{+L/2} \frac{\lambda dz}{\sqrt{z^2 + y^2}}$$

$$\left(r = \sqrt{z^2 + y^2} \right)$$

$$V = \frac{\lambda}{4\pi\epsilon_0} \int_{-L/2}^{+L/2} \frac{dz}{\sqrt{z^2 + y^2}}$$

$$\left(\ln z + \sqrt{z^2 + y^2} \right)$$

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[z + \sqrt{z^2 + y^2} \right]_{-L/2}^{+L/2}$$

$$\left(\left(\frac{-L}{2} \right) = \frac{L^2}{4} \right)$$

$$\left(\ln a - \ln b = \ln \left(\frac{a}{b} \right) \right)$$

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L}{2} + \sqrt{\frac{L^2}{4} + y^2} \right] - \ln \left[-\frac{L}{2} + \sqrt{\frac{L^2}{4} + y^2} \right]$$

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L/2 + \sqrt{\frac{L^2}{4} + y^2}}{-L/2 + \sqrt{\frac{L^2}{4} + y^2}} \right]$$

→ E.P due to line of charge.

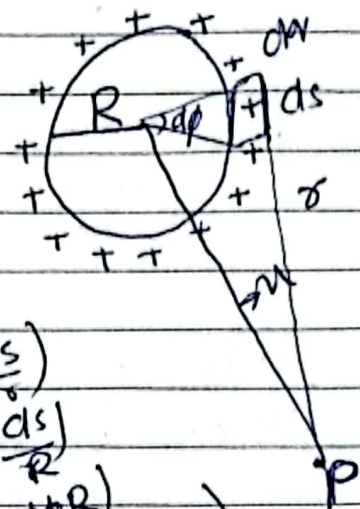
This is the derivation for E.P. at Point P due to line of charge.

↳ depends on (length $\frac{L}{2}$) (y) (density λ)

↳ Resultive component is (Scalar E.P. $\rightarrow$) direct relation with density λ

Electric Potential due to Ring of charge:

Let us consider a (+ve) charge '+q' is uniformly distributed over a ring of radius 'R'. We want to calculate E.P at a point 'P' which is at distance 'x' from centre of ring.



$$\left(\theta = \frac{s}{R} \right)$$

$$\left(ds = R d\theta \right)$$

$$\left(r = \sqrt{x^2 + R^2} \right)$$

Consider a small charge ' dq ' such that linear charge density will be $\lambda = \frac{dq}{ds} \Rightarrow dq = \lambda ds$

First we will calculate small charge dq .

E.P due to small charge ' dq '

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

E.P for entire Ring:

integrate it

$$\int dV = \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{dq}{r}$$

Limit $\int_0^{2\pi}$ For ring of charge

$$V = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\lambda ds}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\lambda R d\phi}{\sqrt{x^2 + R^2}}$$

$$[ds = R d\phi]$$

$$[r = \sqrt{x^2 + R^2}]$$

$$V = \frac{\lambda R}{4\pi\epsilon_0} \times \frac{1}{\sqrt{x^2 + R^2}} \int_0^{2\pi} d\phi$$

$$V = \frac{\lambda R}{4\pi\epsilon_0 \sqrt{x^2 + R^2}} \left[\phi \right]_0^{2\pi}$$

integrate
apply
change
constant

$$V = \frac{\lambda R}{4\pi\epsilon_0 \sqrt{x^2 + R^2}} (2\pi - 0)$$

$$V = \frac{2\pi\lambda R}{4\pi\epsilon_0 (\sqrt{x^2 + R^2})}$$

$2\pi R = \text{Circumference}$
 $\lambda = \text{Charge density}$
 $\rightarrow \text{Total charge} = q$

$$V = \frac{q}{4\pi\epsilon_0 \sqrt{x^2 + R^2}}$$

Electric Potential due to Ring of Charge.

(depends on x and R)

Graph:

(maximum Potential)

at $x=0$ which is at centre of ring.

Potential will be maximum

$$V_{\max} = \frac{q}{4\pi\epsilon_0 \sqrt{3} R^2}$$

$$V_{\max} = \frac{q}{4\pi\epsilon_0 \sqrt{R^2}}$$

$$V_{\max} = \frac{q}{4\pi\epsilon_0 R}$$

minimum Potential:

when $r = \infty$, $V = 0$

$$V_{\min} = \frac{q}{4\pi\epsilon_0 \sqrt{\infty^2 + R^2}}$$

$$V_{\min} = \frac{q}{\infty}$$

$$V_{\min} = 0$$

